

Turing Machines

CSE 211 (Theory of Computation)

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Adapted from slides by
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Programming Techniques for Turing Machines

- A TM is exactly as powerful as a conventional computer.
- We will extend the basic model of the TM by adding features
- None of these tricks extend the basic model of the TM they are only notational conveniences

Storage in the State

- We can use the finite control not only to represent a position in the “program” of the Turing machine, but to hold a finite amount of data.
- Figure suggests this technique.
- This also suggests the idea of multiple tracks.

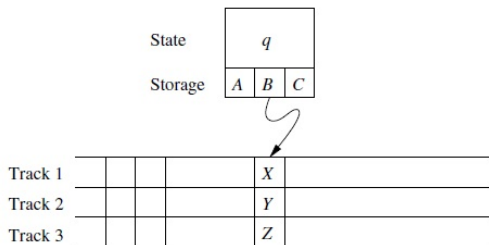
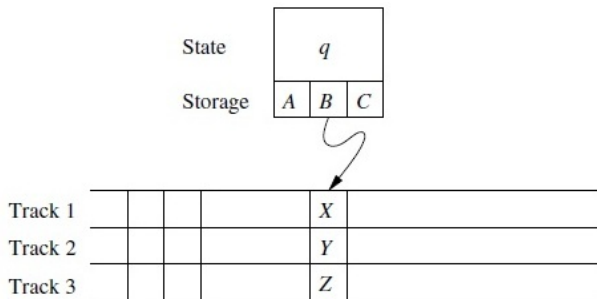


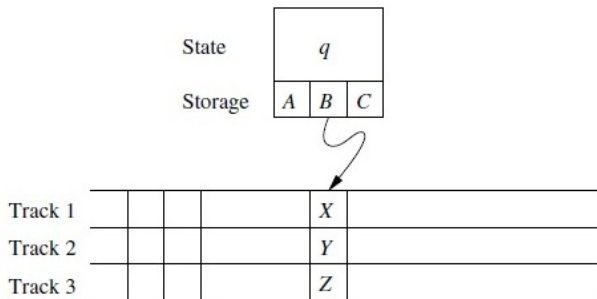
Figure 8.13: A Turing machine viewed as having finite-control storage and multiple tracks

Storage in the State



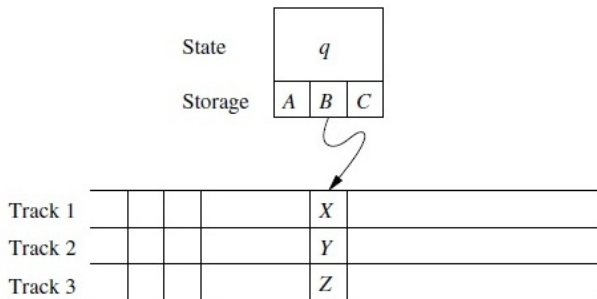
- We see the finite control consisting of not only a “control” state q , but three data elements A , B , and C .
- The technique requires no extension to the TM model.

Storage in the State



- We merely think of the state as a tuple.
- In the case of the diagram, we should think of the state as $[q, A, B, C]$.

Storage in the State



- Regarding states this way allows us to describe transitions in a more systematic way.
- This often makes the strategy behind the TM program more transparent.

Example

- We shall design a TM

$$M = (Q, \{0, 1\}, \{0, 1, B\}, \delta, [q_0, B], \{[q_1, B]\})$$

that remembers in its finite control the first symbol (0 or 1) that it sees.

- And checks that it does not appear elsewhere on its input.
- Thus, M accepts the language $01^* + 10^*$.
- Accepting regular languages such as this one does not stress the ability of Turing machines.
- But it will serve as a simple demonstration.

Example-continued

- The set of states Q is $\{q_0, q_1\} \times \{0, 1, B\}$.
- That is, the states may be thought of as pairs with two components:
 - a) A control portion, q_0 or q_1 .
 - This remembers what the TM is doing.
 - Control state q_0 indicates that M has not yet read its first symbol.
 - q_1 indicates that it has read the symbol.
 - In q_1 it is checking that it does not appear elsewhere, by moving right and hoping to reach a blank cell.
 - b) A data portion.
 - This remembers the first symbol seen, which must be 0 or 1.
 - The symbol B in this component means that no symbol has been read.

Example-continued

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 - Control state q_0 indicates that M has not yet read its first symbol.
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 - b) A data portion.
 - This remembers the first symbol seen, which must be 0 or 1.
 - The symbol B in this component means that no symbol has been read.

Example-continued

The transition function δ of M is as follows:

1. $\delta([q_0, B], a) = ([q_1, a], a, R)$ for $a = 0$ or $a = 1$.
 - Initially, q_0 is the control state, and the data portion of the state is B .
 - The symbol scanned is copied into the second component of the state.
 - And M moves right, entering control state q_1 as it does so.

Example-continued

The transition function δ of M is as follows:

2. $\delta([q_1, a], \bar{a}) = ([q_1, a], \bar{a}, R)$.
 - Here $\bar{a}$ is the “complement” of a .
 - That is, 0 if $a = 1$ and 1 if $a = 0$.
 - In state q_1 M skips over each symbol 0 or 1 that is different from the one it has stored in its state, and continues moving right.

Example-continued

The transition function δ of M is as follows:

3. $\delta([q_1, a], B) = ([q_1, B], B, R)$ for $a = 0$ or $a = 1$.
 - If M reaches the first blank, it enters the accepting state $[q_1, B]$.

Example-continued

1. $\delta ([q_0, B], a) = ([q_1, a], a, R)$ for $a = 0$ or $a = 1$.
 2. $\delta ([q_1, a], \bar{a}) = ([q_1, a], \bar{a}, R)$.
 3. $\delta ([q_1, a], B) = ([q_1, B], B, R)$ for $a = 0$ or $a = 1$.
-
- Notice that M has no definition for $\delta ([q_1, a], a)$ for $a = 0$ or $a = 1$.
 - Thus, if M encounters a second occurrence of the symbol it stored initially in its finite control, it halts without having entered the accepting state.

Example-continued

$$M = (Q, \{0, 1\}, \{0, 1, B\}, \delta, [q_0, B], \{[q_1, B]\})$$

01111

1. $\delta([q_0, B], a) = ([q_1, a], a, R)$ for $a = 0$ or $a = 1$
2. $\delta([q_1, a], \bar{a}) = ([q_1, a], \bar{a}, R)$
3. $\delta([q_1, a], B) = ([q_1, B], B, R)$ for $a = 0$ or $a = 1$

$[q_0, B]01111$

$\vdash 0[q_1, 0]1111$

$\vdash 01[q_1, 0]111$

$\vdash 011[q_1, 0]11$

$\vdash 0111[q_1, 0]1$

$\vdash 01111[q_1, 0]B$

$\vdash 01111B[q_1, B]B$

Thus M reaches its accepting state and the string is accepted.

Example-continued

$$M = (Q, \{0, 1\}, \{0, 1, B\}, \delta, [q_0, B], \{[q_1, B]\})$$

01111

1. $\delta([q_0, B], a) = ([q_1, a], a, R)$ for $a = 0$ or $a = 1$
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$[q_0, B]01111$

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$\vdash 0111[q_1, 0]1$

$\vdash 01111[q_1, 0]B$

$\vdash 01111B[q_1, B]B$

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Example-continued

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3. $\delta([q_1, a], B) = ([q_1, B], B, R)$ for $a = 0$ or $a = 1$

$[q_0, B]$ 01111

$\vdash 0[q_1, 0] 1111$

$\vdash 01[q_1, 0] 111$

$\vdash 011[q_1, 0] 11$

$\vdash 0111[q_1, 0] 1$

$\vdash 01111[q_1, 0] B$

$\vdash 01111B[q_1, B] B$

Thus M reaches its accepting state and the string is accepted.

Example-continued

$$M = (Q, \{0, 1\}, \{0, 1, B\}, \delta, [q_0, B], \{[q_1, B]\})$$

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3. $\delta([q_1, a], B) = ([q_1, B], B, R)$ for $a = 0$ or $a = 1$

$[q_0, B] 01111$

$\vdash 0[q_1, 0] 1111$

$\vdash 01[q_1, 0] 111$

$\vdash 011[q_1, 0] 11$

$\vdash 0111[q_1, 0] 1$

$\vdash 01111[q_1, 0] B$

$\vdash 01111B[q_1, B] B$

Thus M reaches its accepting state and the string is accepted.

Example-continued

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$[q_0, B] 01111$

$\vdash 0[q_1, 0] 1111$

$\vdash 01[q_1, 0] 111$

$\vdash 011[q_1, 0] 11$

$\vdash 0111[q_1, 0] 1$

$\vdash 01111[q_1, 0] B$

$\vdash 01111B[q_1, B] B$

Thus M reaches its accepting state and the string is accepted.

Example-continued

$$M = (Q, \{0, 1\}, \{0, 1, B\}, \delta, [q_0, B], \{[q_1, B]\})$$

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$[q_0, B] 01111$

$\vdash 0[q_1, 0] 1111$

$\vdash 01[q_1, 0] 111$

$\vdash 011[q_1, 0] 11$

$\vdash 0111[q_1, 0] 1$

$\vdash 01111[q_1, 0] B$

$\vdash 01111B[q_1, B] B$

Thus M reaches its accepting state and the string is accepted.

Example-continued

$$M = (Q, \{0, 1\}, \{0, 1, B\}, \delta, [q_0, B], \{[q_1, B]\})$$

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$[q_0, B]$ 01111

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$\vdash 0111[q_1, 0] 1$

$\vdash 01111[q_1, 0] B$

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Thus M reaches its accepting state and the string is accepted.

Example-continued

$$M = (Q, \{0, 1\}, \{0, 1, B\}, \delta, [q_0, B], \{[q_1, B]\})$$

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$\vdash 01111[q_1, 0] B$

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Thus M reaches its accepting state and the string is accepted.

Example-continued

$$M = (Q, \{0, 1\}, \{0, 1, B\}, \delta, [q_0, B], \{[q_1, B]\})$$

We consider a string which does not belong to the language.

01101

1. $\delta([q_0, B], a) = ([q_1, a], a, R)$ for $a = 0$ or $a = 1$
2. $\delta([q_1, a], \bar{a}) = ([q_1, a], \bar{a}, R)$
3. $\delta([q_1, a], B) = ([q_1, B], B, R)$ for $a = 0$ or $a = 1$

$[q_0, B]01101$

$\vdash 0[q_1, 0]1101$

$\vdash 01[q_1, 0]101$

$\vdash 011[q_1, 0]01$

Thus M it halts without having entered the accepting state.

Example-continued

$$M = (Q, \{0, 1\}, \{0, 1, B\}, \delta, [q_0, B], \{[q_1, B]\})$$

We consider a string which does not belong to the language.

01101

1. $\delta([q_0, B], a) = ([q_1, a], a, R)$ for $a = 0$ or $a = 1$
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3. $\delta([q_1, a], B) = ([q_1, B], B, R)$ for $a = 0$ or $a = 1$

$[q_0, B] 01101$

$\vdash 0[q_1, 0] 1101$

$\vdash 01[q_1, 0] 101$

$\vdash 011[q_1, 0] 01$

Thus M it halts without having entered the accepting state.

Example-continued

$$M = (Q, \{0, 1\}, \{0, 1, B\}, \delta, [q_0, B], \{[q_1, B]\})$$

We consider a string which does not belong to the language.

01101

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$[q_0, B] 01101$

$\vdash 0[q_1, 0] 1101$

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Thus M it halts without having entered the accepting state.

Example-continued

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We consider a string which does not belong to the language.

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3. $\delta([q_1, a], B) = ([q_1, B], B, R)$ for $a = 0$ or $a = 1$

$[q_0, B] 01101$

$\vdash 0[q_1, 0] 1101$

$\vdash 01[q_1, 0] 101$

$\vdash 011[q_1, 0] 01$

Thus M it halts without having entered the accepting state.

Example-continued

$$M = (Q, \{0, 1\}, \{0, 1, B\}, \delta, [q_0, B], \{[q_1, B]\})$$

We consider a string which does not belong to the language.

01101

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3. $\delta([q_1, a], B) = ([q_1, B], B, R)$ for $a = 0$ or $a = 1$

$[q_0, B] 01101$

$\vdash 0[q_1, 0] 1101$

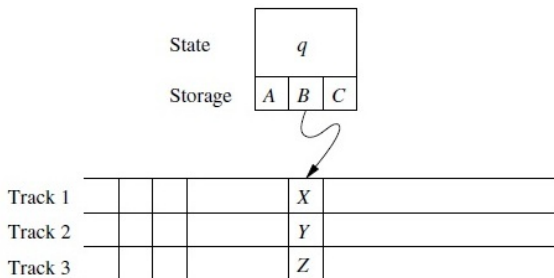
$\vdash 01[q_1, 0] 101$

$\vdash 011[q_1, 0] 01$

Thus M it halts without having entered the accepting state.

Multiple Tracks

- Another useful “trick” is to think of the tape of a Turing machine as composed of several tracks.
- Each track can hold one symbol.
- The tape alphabet of the TM consists of tuples, with one component for each “track.”
- Thus, for instance, the cell scanned by the tape head in figure contains the symbol $[X, Y, Z]$.



Multiple Tracks

- Like the technique of storage in the finite control, using multiple tracks does not extend what the Turing machine can do.
- It is simply a way to view tape symbols and to imagine that they have a useful structure.

Example

- A common use of multiple tracks is to treat one track as holding the data and a second track as holding a mark.
- We can check off each symbol as we “use” it.
- Or we can keep track of a small number of positions within the data by marking only those positions.
- Previous examples were instances of this technique.
- But in those examples we did not think explicitly of the tape as if it were composed of tracks.

Example-continued

- In the present example, we shall use a second track explicitly to recognize the non-context-free language

$$L_{wcw} = \{wcw \mid w \text{ is in } (0 + 1)^+\}$$

Example-continued

0	1	1	c	0	1	1

Example-continued

✓						
0	1	1	c	0	1	1

Example-continued

✓				✓		
0	1	1	c	0	1	1

Example-continued

✓	✓			✓		
0	1	1	c	0	1	1

Example-continued

✓	✓			✓	✓	
0	1	1	c	0	1	1

Example-continued

✓	✓	✓		✓	✓	
0	1	1	c	0	1	1

Example-continued

✓	✓	✓		✓	✓	✓
0	1	1	c	0	1	1

Example-continued

- The Turing machine we shall design is:

$$M = (Q, \Sigma, \Gamma, \delta, [q_1, B], [B, B], \{[q_9, B]\})$$

Q: The set of states is $\{q_1, q_2, \dots, q_9\} \times \{0, 1, B\}$.

- That is, pairs consisting of a control state q_i and a data component: 0, 1, or blank.
- We again use the technique of storage in the finite control, as we allow the state to remember an input symbol 0 or 1.

Example-continued

- The Turing machine we shall design is:

$$M = (Q, \Sigma, \Gamma, \delta, [q_1, B], [B, B], \{[q_9, B]\})$$

⌈: The set of tape symbols is $\{B, *\} \times \{0, 1, c, B\}$.

- The first component, or track, can be either blank or “checked,” represented by the symbols B and $*$, respectively.
- We use the $*$ to check off symbols of the first and second groups of 0’s and 1’s.
- This eventually confirms that the string to the left of the center marker c is the same as the string to its right.
- The second component of the tape symbol is what we think of as the tape symbol itself.
- That is, we may think of the symbol $[B, X]$ as if it were the tape symbol X , for $X = 0, 1, c, B$.

Example-continued

- The Turing machine we shall design is:

$$M = (Q, \Sigma, \Gamma, \delta, [q_1, B], [B, B], \{[q_9, B]\})$$

Σ : The input symbols are $[B, 0]$, $[B, 1]$ and $[B, c]$.

- This, as mentioned above, we identify with 0, 1 and c, respectively.

Example-continued

- The Turing machine we shall design is:

$$M = (Q, \Sigma, \Gamma, \delta, [q_1, B], [B, B], \{[q_9, B]\})$$

δ : The transition function.

Example-continued

δ is defined by the following rules, in which a and b each may stand for either 0 or 1.

1. $\delta ([q_1, B], [B, a]) = ([q_2, a], [* , a], R)$.
 - In the initial state,
 - M picks up the symbol a (which can be either 0 or 1),
 - stores it in its finite control,
 - goes to control state q_2 ,
 - “checks off” the symbol it just scanned,
 - and moves right.
 - Notice that by changing the first component of the tape symbol from B to $*$, it performs the check-off.

Example-continued

δ is defined by the following rules, in which a and b each may stand for either 0 or 1.

2. $\delta ([q_2, a], [B, b]) = ([q_2, a], [B, b], R)$.
 - M moves right, looking for the symbol c .
 - Remember that a and b can each be either 0 or 1, independently, but can not be c .

Example-continued

δ is defined by the following rules, in which a and b each may stand for either 0 or 1.

3. $\delta ([q_2, a], [B, c]) = ([q_3, a], [B, c], R)$.
 - When M finds the c , it continues to move right, but changes to control state q_3 .

Example-continued

δ is defined by the following rules, in which a and b each may stand for either 0 or 1.

4. $\delta ([q_3, a], [* , b]) = ([q_3, a], [* , b], R)$.
 - In state q_3 , M continues past all checked symbols.

Example-continued

δ is defined by the following rules, in which a and b each may stand for either 0 or 1.

5. $\delta ([q_3, a], [B, a]) = ([q_4, B], [*, a], L)$.
- If the first unchecked symbol that M finds is the same as the symbol in its finite control, it checks this symbol.
 - It has matched the corresponding symbol from the first block of 0's and 1's.
 - M goes to control state q_4 , dropping the symbol from its finite control, and starts moving left.

Example-continued

δ is defined by the following rules, in which a and b each may stand for either 0 or 1.

6. $\delta ([q_4, B], [* , a]) = ([q_4, B], [* , a], L)$.
 - M moves left over checked symbols.

Example-continued

δ is defined by the following rules, in which a and b each may stand for either 0 or 1.

7. $\delta ([q_4, B], [B, c]) = ([q_5, B], [B, c], L)$.
- When M encounters the symbol c , it switches to state q_5 and continues left.
 - In state q_5 , M must make a decision, depending on whether or not the symbol immediately to the left of the c is checked or unchecked.
 - If checked, then we have already considered the entire first block of 0's and 1's – those to the left of the c .
 - We must make sure that all the 0's and 1's to the right of the c are also checked.

Example-continued

δ is defined by the following rules, in which a and b each may stand for either 0 or 1.

7 (cont). $\delta ([q_4, B], [B, c]) = ([q_5, B], [B, c], L)$.

- We accept if no unchecked symbols remain to the right of the c .
- If the symbol immediately to the left of the c is unchecked,
 - we find the leftmost unchecked symbol,
 - pick it up, and
 - start the cycle that began in state q_1 .

Example-continued

δ is defined by the following rules, in which a and b each may stand for either 0 or 1.

8. $\delta ([q_5, B], [B, a]) = ([q_6, B], [B, a], L)$.
- This branch covers the case where the symbol to the left of c is unchecked.
 - M goes to state q_6 and continues left, looking for a checked symbol.

Example-continued

δ is defined by the following rules, in which a and b each may stand for either 0 or 1.

9. $\delta ([q_6, B], [B, a]) = ([q_6, B], [B, a], L)$.
 - As long as symbols are unchecked, M remains in state q_6 and proceeds left.

Example-continued

δ is defined by the following rules, in which a and b each may stand for either 0 or 1.

10. $\delta ([q_6, B], [*, a]) = ([q_1, B], [*, a], R)$.
- When the checked symbol is found, M enters state q_1 and moves right to pick up the first unchecked symbol.

Example-continued

δ is defined by the following rules, in which a and b each may stand for either 0 or 1.

11. $\delta ([q_5, B], [*, a]) = ([q_7, B], [*, a], R)$.

- Now, let us pick up the branch from state q_5 .
- In q_5 we have just moved left from the c and find a checked symbol.
- We start moving right again, entering state q_7 .

Example-continued

δ is defined by the following rules, in which a and b each may stand for either 0 or 1.

12. $\delta ([q_7, B], [B, c]) = ([q_8, B], [B, c], R)$.

- In state q_7 we shall surely see the c .
- We enter state q_8 as we do so, and proceed right.

Example-continued

δ is defined by the following rules, in which a and b each may stand for either 0 or 1.

13. $\delta ([q_8, B], [* , a]) = ([q_8, B], [* , a], R)$.
- M moves right in state q_8 , skipping over any checked 0's or 1's that it finds.

Example-continued

δ is defined by the following rules, in which a and b each may stand for either 0 or 1.

14. $\delta ([q_8, B], [B, B]) = ([q_9, B], [B, B], R)$.

- If M reaches a blank cell in state q_8 without encountering any unchecked 0 or 1, then M accepts.
- If M first finds an unchecked 0 or 1, then the blocks before and after the c do not match, and M halts without accepting.

1. $\delta([q_1, B], [B, a]) = ([q_2, a], [*, a], R)$
2. $\delta([q_2, a], [B, b]) = ([q_2, a], [B, b], R)$
3. $\delta([q_2, a], [B, c]) = ([q_3, a], [B, c], R)$
4. $\delta([q_3, a], [*, b]) = ([q_3, a], [*, b], R)$
5. $\delta([q_3, a], [B, a]) = ([q_4, B], [*, a], L)$
6. $\delta([q_4, B], [*, a]) = ([q_4, B], [*, a], L)$
7. $\delta([q_4, B], [B, c]) = ([q_5, B], [B, c], L)$
8. $\delta([q_5, B], [B, a]) = ([q_6, B], [B, a], L)$
9. $\delta([q_6, B], [B, a]) = ([q_6, B], [B, a], L)$
10. $\delta([q_6, B], [*, a]) = ([q_1, B], [*, a], R)$
11. $\delta([q_5, B], [*, a]) = ([q_7, B], [*, a], R)$
12. $\delta([q_7, B], [B, c]) = ([q_8, B], [B, c], R)$
13. $\delta([q_8, B], [*, a]) = ([q_8, B], [*, a], R)$
14. $\delta([q_8, B], [B, B]) = ([q_9, B], [B, B], R)$

Before

State: $[q_1, B]$

B	B	B	B	B	B	B
0	1	1	c	0	1	1

M picks up the symbol 0, stores it in its finite control, goes to control state q_2 , “checks off” the symbol just scanned, and moves right.

After

State: $[q_2, 0]$

*	B	B	B	B	B	B
0	1	1	c	0	1	1

1. $\delta([q_1, B], [B, a]) = ([q_2, a], [*, a], R)$
2. $\delta([q_2, a], [B, b]) = ([q_2, a], [B, b], R)$
3. $\delta([q_2, a], [B, c]) = ([q_3, a], [B, c], R)$
4. $\delta([q_3, a], [*, b]) = ([q_3, a], [*, b], R)$
5. $\delta([q_3, a], [B, a]) = ([q_4, B], [*, a], L)$
6. $\delta([q_4, B], [*, a]) = ([q_4, B], [*, a], L)$
7. $\delta([q_4, B], [B, c]) = ([q_5, B], [B, c], L)$
8. $\delta([q_5, B], [B, a]) = ([q_6, B], [B, a], L)$
9. $\delta([q_6, B], [B, a]) = ([q_6, B], [B, a], L)$
10. $\delta([q_6, B], [*, a]) = ([q_1, B], [*, a], R)$
11. $\delta([q_5, B], [*, a]) = ([q_7, B], [*, a], R)$
12. $\delta([q_7, B], [B, c]) = ([q_8, B], [B, c], R)$
13. $\delta([q_8, B], [*, a]) = ([q_8, B], [*, a], R)$
14. $\delta([q_8, B], [B, B]) = ([q_9, B], [B, B], R)$

Before

State: $[q_2, 0]$

*	B	B	B	B	B	B
0	1	1	c	0	1	1

M moves right, looking for the symbol c .

After

State: $[q_2, 0]$

*	B	B	B	B	B	B
0	1	1	c	0	1	1

1. $\delta([q_1, B], [B, a]) = ([q_2, a], [*, a], R)$
2. $\delta([q_2, a], [B, b]) = ([q_2, a], [B, b], R)$
3. $\delta([q_2, a], [B, c]) = ([q_3, a], [B, c], R)$
4. $\delta([q_3, a], [*, b]) = ([q_3, a], [*, b], R)$
5. $\delta([q_3, a], [B, a]) = ([q_4, B], [*, a], L)$
6. $\delta([q_4, B], [*, a]) = ([q_4, B], [*, a], L)$
7. $\delta([q_4, B], [B, c]) = ([q_5, B], [B, c], L)$
8. $\delta([q_5, B], [B, a]) = ([q_6, B], [B, a], L)$
9. $\delta([q_6, B], [B, a]) = ([q_6, B], [B, a], L)$
10. $\delta([q_6, B], [*, a]) = ([q_1, B], [*, a], R)$
11. $\delta([q_5, B], [*, a]) = ([q_7, B], [*, a], R)$
12. $\delta([q_7, B], [B, c]) = ([q_8, B], [B, c], R)$
13. $\delta([q_8, B], [*, a]) = ([q_8, B], [*, a], R)$
14. $\delta([q_8, B], [B, B]) = ([q_9, B], [B, B], R)$

Before

State: $[q_2, 0]$

*	B	B	B	B	B	B
0	1	1	c	0	1	1

M moves right, looking for the symbol c .

After

State: $[q_2, 0]$

*	B	B	B	B	B	B
0	1	1	c	0	1	1

Before

State: $[q_2, 0]$

*	B	B	B	B	B	B
0	1	1	c	0	1	1

When M finds the c , it continues to move right, but changes to control state q_3

After

State: $[q_3, 0]$

*	B	B	B	B	B	B
0	1	1	c	0	1	1

1. $\delta([q_1, B], [B, a]) = ([q_2, a], [*, a], R)$
2. $\delta([q_2, a], [B, b]) = ([q_2, a], [B, b], R)$
3. $\delta([q_2, a], [B, c]) = ([q_3, a], [B, c], R)$
4. $\delta([q_3, a], [*, b]) = ([q_3, a], [*, b], R)$
5. $\delta([q_3, a], [B, a]) = ([q_4, B], [*, a], L)$
6. $\delta([q_4, B], [*, a]) = ([q_4, B], [*, a], L)$
7. $\delta([q_4, B], [B, c]) = ([q_5, B], [B, c], L)$
8. $\delta([q_5, B], [B, a]) = ([q_6, B], [B, a], L)$
9. $\delta([q_6, B], [B, a]) = ([q_6, B], [B, a], L)$
10. $\delta([q_6, B], [*, a]) = ([q_1, B], [*, a], R)$
11. $\delta([q_5, B], [*, a]) = ([q_7, B], [*, a], R)$
12. $\delta([q_7, B], [B, c]) = ([q_8, B], [B, c], R)$
13. $\delta([q_8, B], [*, a]) = ([q_8, B], [*, a], R)$
14. $\delta([q_8, B], [B, B]) = ([q_9, B], [B, B], R)$

Before

State: $[q_3, 0]$

*	B	B	B	B	B	B
0	1	1	c	0	1	1

1. $\delta([q_1, B], [B, a]) = ([q_2, a], [*, a], R)$
2. $\delta([q_2, a], [B, b]) = ([q_2, a], [B, b], R)$
3. $\delta([q_2, a], [B, c]) = ([q_3, a], [B, c], R)$
4. $\delta([q_3, a], [*, b]) = ([q_3, a], [*, b], R)$
5. $\delta([q_3, a], [B, a]) = ([q_4, B], [*, a], L)$
6. $\delta([q_4, B], [*, a]) = ([q_4, B], [*, a], L)$
7. $\delta([q_4, B], [B, c]) = ([q_5, B], [B, c], L)$
8. $\delta([q_5, B], [B, a]) = ([q_6, B], [B, a], L)$
9. $\delta([q_6, B], [B, a]) = ([q_6, B], [B, a], L)$
10. $\delta([q_6, B], [*, a]) = ([q_1, B], [*, a], R)$
11. $\delta([q_5, B], [*, a]) = ([q_7, B], [*, a], R)$
12. $\delta([q_7, B], [B, c]) = ([q_8, B], [B, c], R)$
13. $\delta([q_8, B], [*, a]) = ([q_8, B], [*, a], R)$
14. $\delta([q_8, B], [B, B]) = ([q_9, B], [B, B], R)$

Matched the corresponding symbol from first block. Checks this. Goes to control state q_4 . Drops the symbol from finite control. Starts moving left

After

State: $[q_4, B]$

*	B	B	B	*	B	B
0	1	1	c	0	1	1

Before

State: $[q_4, B]$

*	B	B	B	*	B	B
0	1	1	c	0	1	1

When M encounters the symbol c , it switches to state q_5 and continues left

After

State: $[q_5, B]$

*	B	B	B	*	B	B
0	1	1	c	0	1	1

1. $\delta([q_1, B], [B, a]) = ([q_2, a], [*, a], R)$
2. $\delta([q_2, a], [B, b]) = ([q_2, a], [B, b], R)$
3. $\delta([q_2, a], [B, c]) = ([q_3, a], [B, c], R)$
4. $\delta([q_3, a], [*, b]) = ([q_3, a], [*, b], R)$
5. $\delta([q_3, a], [B, a]) = ([q_4, B], [*, a], L)$
6. $\delta([q_4, B], [*, a]) = ([q_4, B], [*, a], L)$
- 7. $\delta([q_4, B], [B, c]) = ([q_5, B], [B, c], L)$**
8. $\delta([q_5, B], [B, a]) = ([q_6, B], [B, a], L)$
9. $\delta([q_6, B], [B, a]) = ([q_6, B], [B, a], L)$
10. $\delta([q_6, B], [*, a]) = ([q_1, B], [*, a], R)$
11. $\delta([q_5, B], [*, a]) = ([q_7, B], [*, a], R)$
12. $\delta([q_7, B], [B, c]) = ([q_8, B], [B, c], R)$
13. $\delta([q_8, B], [*, a]) = ([q_8, B], [*, a], R)$
14. $\delta([q_8, B], [B, B]) = ([q_9, B], [B, B], R)$

1. $\delta([q_1, B], [B, a]) = ([q_2, a], [*, a], R)$
2. $\delta([q_2, a], [B, b]) = ([q_2, a], [B, b], R)$
3. $\delta([q_2, a], [B, c]) = ([q_3, a], [B, c], R)$
4. $\delta([q_3, a], [*, b]) = ([q_3, a], [*, b], R)$
5. $\delta([q_3, a], [B, a]) = ([q_4, B], [*, a], L)$
6. $\delta([q_4, B], [*, a]) = ([q_4, B], [*, a], L)$
7. $\delta([q_4, B], [B, c]) = ([q_5, B], [B, c], L)$
- 8. $\delta([q_5, B], [B, a]) = ([q_6, B], [B, a], L)$**
9. $\delta([q_6, B], [B, a]) = ([q_6, B], [B, a], L)$
10. $\delta([q_6, B], [*, a]) = ([q_1, B], [*, a], R)$
11. $\delta([q_5, B], [*, a]) = ([q_7, B], [*, a], R)$
12. $\delta([q_7, B], [B, c]) = ([q_8, B], [B, c], R)$
13. $\delta([q_8, B], [*, a]) = ([q_8, B], [*, a], R)$
14. $\delta([q_8, B], [B, B]) = ([q_9, B], [B, B], R)$

Before

State: $[q_5, B]$

*	B	B	B	*	B	B
0	1	1	c	0	1	1

This branch covers the case where the symbol to the left of c is unchecked. M goes to state q_6 and continues left, looking for a checked symbol

After

State: $[q_6, B]$

*	B	B	B	*	B	B
0	1	1	c	0	1	1

1. $\delta([q_1, B], [B, a]) = ([q_2, a], [*, a], R)$
2. $\delta([q_2, a], [B, b]) = ([q_2, a], [B, b], R)$
3. $\delta([q_2, a], [B, c]) = ([q_3, a], [B, c], R)$
4. $\delta([q_3, a], [*, b]) = ([q_3, a], [*, b], R)$
5. $\delta([q_3, a], [B, a]) = ([q_4, B], [*, a], L)$
6. $\delta([q_4, B], [*, a]) = ([q_4, B], [*, a], L)$
7. $\delta([q_4, B], [B, c]) = ([q_5, B], [B, c], L)$
8. $\delta([q_5, B], [B, a]) = ([q_6, B], [B, a], L)$
- 9. $\delta([q_6, B], [B, a]) = ([q_6, B], [B, a], L)$**
10. $\delta([q_6, B], [*, a]) = ([q_1, B], [*, a], R)$
11. $\delta([q_5, B], [*, a]) = ([q_7, B], [*, a], R)$
12. $\delta([q_7, B], [B, c]) = ([q_8, B], [B, c], R)$
13. $\delta([q_8, B], [*, a]) = ([q_8, B], [*, a], R)$
14. $\delta([q_8, B], [B, B]) = ([q_9, B], [B, B], R)$

Before

State: $[q_6, B]$

*	B	B	B	*	B	B
0	1	1	c	0	1	1

As long as symbols are unchecked,
M remains in state q_6 and proceeds left

After

State: $[q_6, B]$

*	B	B	B	*	B	B
0	1	1	c	0	1	1

Before

State: $[q_6, B]$

*	B	B	B	*	B	B
0	1	1	c	0	1	1

When the checked symbol is found, M enters state q_1 and moves right to pick up the first unchecked symbol

After

State: $[q_1, B]$

*	B	B	B	*	B	B
0	1	1	c	0	1	1

1. $\delta([q_1, B], [B, a]) = ([q_2, a], [*, a], R)$
2. $\delta([q_2, a], [B, b]) = ([q_2, a], [B, b], R)$
3. $\delta([q_2, a], [B, c]) = ([q_3, a], [B, c], R)$
4. $\delta([q_3, a], [*, b]) = ([q_3, a], [*, b], R)$
5. $\delta([q_3, a], [B, a]) = ([q_4, B], [*, a], L)$
6. $\delta([q_4, B], [*, a]) = ([q_4, B], [*, a], L)$
7. $\delta([q_4, B], [B, c]) = ([q_5, B], [B, c], L)$
8. $\delta([q_5, B], [B, a]) = ([q_6, B], [B, a], L)$
9. $\delta([q_6, B], [B, a]) = ([q_6, B], [B, a], L)$
10. $\delta([q_6, B], [*, a]) = ([q_1, B], [*, a], R)$
11. $\delta([q_5, B], [*, a]) = ([q_7, B], [*, a], R)$
12. $\delta([q_7, B], [B, c]) = ([q_8, B], [B, c], R)$
13. $\delta([q_8, B], [*, a]) = ([q_8, B], [*, a], R)$
14. $\delta([q_8, B], [B, B]) = ([q_9, B], [B, B], R)$

1. $\delta([q_1, B], [B, a]) = ([q_2, a], [*, a], R)$
2. $\delta([q_2, a], [B, b]) = ([q_2, a], [B, b], R)$
3. $\delta([q_2, a], [B, c]) = ([q_3, a], [B, c], R)$
4. $\delta([q_3, a], [*, b]) = ([q_3, a], [*, b], R)$
5. $\delta([q_3, a], [B, a]) = ([q_4, B], [*, a], L)$
6. $\delta([q_4, B], [*, a]) = ([q_4, B], [*, a], L)$
7. $\delta([q_4, B], [B, c]) = ([q_5, B], [B, c], L)$
8. $\delta([q_5, B], [B, a]) = ([q_6, B], [B, a], L)$
9. $\delta([q_6, B], [B, a]) = ([q_6, B], [B, a], L)$
10. $\delta([q_6, B], [*, a]) = ([q_1, B], [*, a], R)$
11. $\delta([q_5, B], [*, a]) = ([q_7, B], [*, a], R)$
12. $\delta([q_7, B], [B, c]) = ([q_8, B], [B, c], R)$
13. $\delta([q_8, B], [*, a]) = ([q_8, B], [*, a], R)$
14. $\delta([q_8, B], [B, B]) = ([q_9, B], [B, B], R)$

Before

State: $[q_1, B]$

*	B	B	B	*	B	B
0	1	1	c	0	1	1

M picks up the symbol 1, stores it in its finite control, goes to control state q_2 , “checks off” the symbol it just scanned, and moves right

After

State: $[q_2, 1]$

*	*	B	B	*	B	B
0	1	1	c	0	1	1

Before

State: $[q_2, 1]$

*	*	B	B	*	B	B
0	1	1	c	0	1	1

 M moves right, looking for the symbol c

After

State: $[q_2, 1]$

*	*	B	B	*	B	B
0	1	1	c	0	1	1

1. $\delta([q_1, B], [B, a]) = ([q_2, a], [*, a], R)$
2. $\delta([q_2, a], [B, b]) = ([q_2, a], [B, b], R)$
3. $\delta([q_2, a], [B, c]) = ([q_3, a], [B, c], R)$
4. $\delta([q_3, a], [*, b]) = ([q_3, a], [*, b], R)$
5. $\delta([q_3, a], [B, a]) = ([q_4, B], [*, a], L)$
6. $\delta([q_4, B], [*, a]) = ([q_4, B], [*, a], L)$
7. $\delta([q_4, B], [B, c]) = ([q_5, B], [B, c], L)$
8. $\delta([q_5, B], [B, a]) = ([q_6, B], [B, a], L)$
9. $\delta([q_6, B], [B, a]) = ([q_6, B], [B, a], L)$
10. $\delta([q_6, B], [*, a]) = ([q_1, B], [*, a], R)$
11. $\delta([q_5, B], [*, a]) = ([q_7, B], [*, a], R)$
12. $\delta([q_7, B], [B, c]) = ([q_8, B], [B, c], R)$
13. $\delta([q_8, B], [*, a]) = ([q_8, B], [*, a], R)$
14. $\delta([q_8, B], [B, B]) = ([q_9, B], [B, B], R)$

Before

State: $[q_2, 1]$

*	*	B	B	*	B	B
0	1	1	c	0	1	1

When M finds the c , it continues to move right, but changes to control state q_3

After

State: $[q_3, 1]$

*	*	B	B	*	B	B
0	1	1	c	0	1	1

1. $\delta([q_1, B], [B, a]) = ([q_2, a], [*, a], R)$
2. $\delta([q_2, a], [B, b]) = ([q_2, a], [B, b], R)$
3. $\delta([q_2, a], [B, c]) = ([q_3, a], [B, c], R)$
4. $\delta([q_3, a], [*, b]) = ([q_3, a], [*, b], R)$
5. $\delta([q_3, a], [B, a]) = ([q_4, B], [*, a], L)$
6. $\delta([q_4, B], [*, a]) = ([q_4, B], [*, a], L)$
7. $\delta([q_4, B], [B, c]) = ([q_5, B], [B, c], L)$
8. $\delta([q_5, B], [B, a]) = ([q_6, B], [B, a], L)$
9. $\delta([q_6, B], [B, a]) = ([q_6, B], [B, a], L)$
10. $\delta([q_6, B], [*, a]) = ([q_1, B], [*, a], R)$
11. $\delta([q_5, B], [*, a]) = ([q_7, B], [*, a], R)$
12. $\delta([q_7, B], [B, c]) = ([q_8, B], [B, c], R)$
13. $\delta([q_8, B], [*, a]) = ([q_8, B], [*, a], R)$
14. $\delta([q_8, B], [B, B]) = ([q_9, B], [B, B], R)$

Before

State: $[q_3, 1]$

*	*	B	B	*	B	B
0	1	1	c	0	1	1

 M continues past all checked symbols

After

State: $[q_3, 1]$

*	*	B	B	*	B	B
0	1	1	c	0	1	1

1. $\delta([q_1, B], [B, a]) = ([q_2, a], [*, a], R)$
2. $\delta([q_2, a], [B, b]) = ([q_2, a], [B, b], R)$
3. $\delta([q_2, a], [B, c]) = ([q_3, a], [B, c], R)$
4. $\delta([q_3, a], [*, b]) = ([q_3, a], [*, b], R)$
5. $\delta([q_3, a], [B, a]) = ([q_4, B], [*, a], L)$
6. $\delta([q_4, B], [*, a]) = ([q_4, B], [*, a], L)$
7. $\delta([q_4, B], [B, c]) = ([q_5, B], [B, c], L)$
8. $\delta([q_5, B], [B, a]) = ([q_6, B], [B, a], L)$
9. $\delta([q_6, B], [B, a]) = ([q_6, B], [B, a], L)$
10. $\delta([q_6, B], [*, a]) = ([q_1, B], [*, a], R)$
11. $\delta([q_5, B], [*, a]) = ([q_7, B], [*, a], R)$
12. $\delta([q_7, B], [B, c]) = ([q_8, B], [B, c], R)$
13. $\delta([q_8, B], [*, a]) = ([q_8, B], [*, a], R)$
14. $\delta([q_8, B], [B, B]) = ([q_9, B], [B, B], R)$

1. $\delta([q_1, B], [B, a]) = ([q_2, a], [*, a], R)$
2. $\delta([q_2, a], [B, b]) = ([q_2, a], [B, b], R)$
3. $\delta([q_2, a], [B, c]) = ([q_3, a], [B, c], R)$
4. $\delta([q_3, a], [*, b]) = ([q_3, a], [*, b], R)$
5. $\delta([q_3, a], [B, a]) = ([q_4, B], [*, a], L)$
6. $\delta([q_4, B], [*, a]) = ([q_4, B], [*, a], L)$
7. $\delta([q_4, B], [B, c]) = ([q_5, B], [B, c], L)$
8. $\delta([q_5, B], [B, a]) = ([q_6, B], [B, a], L)$
9. $\delta([q_6, B], [B, a]) = ([q_6, B], [B, a], L)$
10. $\delta([q_6, B], [*, a]) = ([q_1, B], [*, a], R)$
11. $\delta([q_5, B], [*, a]) = ([q_7, B], [*, a], R)$
12. $\delta([q_7, B], [B, c]) = ([q_8, B], [B, c], R)$
13. $\delta([q_8, B], [*, a]) = ([q_8, B], [*, a], R)$
14. $\delta([q_8, B], [B, B]) = ([q_9, B], [B, B], R)$

Before

State: $[q_3, 1]$

*	*	B	B	*	B	B
0	1	1	c	0	1	1

Matched the corresponding symbol from first block. Checks this. Goes to control state q_4 . Drops the symbol from finite control. Starts moving left

After

State: $[q_4, B]$

*	*	B	B	*	*	B
0	1	1	c	0	1	1

1. $\delta([q_1, B], [B, a]) = ([q_2, a], [*, a], R)$
2. $\delta([q_2, a], [B, b]) = ([q_2, a], [B, b], R)$
3. $\delta([q_2, a], [B, c]) = ([q_3, a], [B, c], R)$
4. $\delta([q_3, a], [*, b]) = ([q_3, a], [*, b], R)$
5. $\delta([q_3, a], [B, a]) = ([q_4, B], [*, a], L)$
6. $\delta([q_4, B], [*, a]) = ([q_4, B], [*, a], L)$
7. $\delta([q_4, B], [B, c]) = ([q_5, B], [B, c], L)$
8. $\delta([q_5, B], [B, a]) = ([q_6, B], [B, a], L)$
9. $\delta([q_6, B], [B, a]) = ([q_6, B], [B, a], L)$
10. $\delta([q_6, B], [*, a]) = ([q_1, B], [*, a], R)$
11. $\delta([q_5, B], [*, a]) = ([q_7, B], [*, a], R)$
12. $\delta([q_7, B], [B, c]) = ([q_8, B], [B, c], R)$
13. $\delta([q_8, B], [*, a]) = ([q_8, B], [*, a], R)$
14. $\delta([q_8, B], [B, B]) = ([q_9, B], [B, B], R)$

Before

State: $[q_4, B]$

*	*	B	B	*	*	B
0	1	1	c	0	1	1

M moves left over checked symbols

After

State: $[q_4, B]$

*	*	B	B	*	*	B
0	1	1	c	0	1	1

1. $\delta([q_1, B], [B, a]) = ([q_2, a], [*, a], R)$
2. $\delta([q_2, a], [B, b]) = ([q_2, a], [B, b], R)$
3. $\delta([q_2, a], [B, c]) = ([q_3, a], [B, c], R)$
4. $\delta([q_3, a], [*, b]) = ([q_3, a], [*, b], R)$
5. $\delta([q_3, a], [B, a]) = ([q_4, B], [*, a], L)$
6. $\delta([q_4, B], [*, a]) = ([q_4, B], [*, a], L)$
- 7. $\delta([q_4, B], [B, c]) = ([q_5, B], [B, c], L)$**
8. $\delta([q_5, B], [B, a]) = ([q_6, B], [B, a], L)$
9. $\delta([q_6, B], [B, a]) = ([q_6, B], [B, a], L)$
10. $\delta([q_6, B], [*, a]) = ([q_1, B], [*, a], R)$
11. $\delta([q_5, B], [*, a]) = ([q_7, B], [*, a], R)$
12. $\delta([q_7, B], [B, c]) = ([q_8, B], [B, c], R)$
13. $\delta([q_8, B], [*, a]) = ([q_8, B], [*, a], R)$
14. $\delta([q_8, B], [B, B]) = ([q_9, B], [B, B], R)$

Before

State: $[q_4, B]$

*	*	B	B	*	*	B
0	1	1	c	0	1	1

Encounters the symbol c , switches to state q_5 and continues left

After

State: $[q_5, B]$

*	*	B	B	*	*	B
0	1	1	c	0	1	1

1. $\delta([q_1, B], [B, a]) = ([q_2, a], [*, a], R)$
2. $\delta([q_2, a], [B, b]) = ([q_2, a], [B, b], R)$
3. $\delta([q_2, a], [B, c]) = ([q_3, a], [B, c], R)$
4. $\delta([q_3, a], [*, b]) = ([q_3, a], [*, b], R)$
5. $\delta([q_3, a], [B, a]) = ([q_4, B], [*, a], L)$
6. $\delta([q_4, B], [*, a]) = ([q_4, B], [*, a], L)$
7. $\delta([q_4, B], [B, c]) = ([q_5, B], [B, c], L)$
8. $\delta([q_5, B], [B, a]) = ([q_6, B], [B, a], L)$
9. $\delta([q_6, B], [B, a]) = ([q_6, B], [B, a], L)$
10. $\delta([q_6, B], [*, a]) = ([q_1, B], [*, a], R)$
11. $\delta([q_5, B], [*, a]) = ([q_7, B], [*, a], R)$
12. $\delta([q_7, B], [B, c]) = ([q_8, B], [B, c], R)$
13. $\delta([q_8, B], [*, a]) = ([q_8, B], [*, a], R)$
14. $\delta([q_8, B], [B, B]) = ([q_9, B], [B, B], R)$

Before

State: $[q_5, B]$

*	*	B	B	*	*	B
0	1	1	c	0	1	1

.....

.....

After

State: $[q_2, 1]$

*	*	B	B	*	*	B
0	1	1	c	0	1	1

Before

State: $[q_2, 1]$

*	*	*	B	*	*	B
0	1	1	c	0	1	1

.....

.....

After

State: $[q_3, 1]$

*	*	*	B	*	*	B
0	1	1	c	0	1	1

1. $\delta([q_1, B], [B, a]) = ([q_2, a], [*, a], R)$
2. $\delta([q_2, a], [B, b]) = ([q_2, a], [B, b], R)$
3. $\delta([q_2, a], [B, c]) = ([q_3, a], [B, c], R)$
4. $\delta([q_3, a], [*, b]) = ([q_3, a], [*, b], R)$
5. $\delta([q_3, a], [B, a]) = ([q_4, B], [*, a], L)$
6. $\delta([q_4, B], [*, a]) = ([q_4, B], [*, a], L)$
7. $\delta([q_4, B], [B, c]) = ([q_5, B], [B, c], L)$
8. $\delta([q_5, B], [B, a]) = ([q_6, B], [B, a], L)$
9. $\delta([q_6, B], [B, a]) = ([q_6, B], [B, a], L)$
10. $\delta([q_6, B], [*, a]) = ([q_1, B], [*, a], R)$
11. $\delta([q_5, B], [*, a]) = ([q_7, B], [*, a], R)$
12. $\delta([q_7, B], [B, c]) = ([q_8, B], [B, c], R)$
13. $\delta([q_8, B], [*, a]) = ([q_8, B], [*, a], R)$
14. $\delta([q_8, B], [B, B]) = ([q_9, B], [B, B], R)$

Before

State: $[q_3, 1]$

*	*	*	B	*	*	B
0	1	1	c	0	1	1

Matched the corresponding symbol from first block. Checks this. Goes to control state q_4 . Drops the symbol from finite control. Starts moving left

After

State: $[q_4, B]$

*	*	*	B	*	*	*
0	1	1	c	0	1	1

1. $\delta([q_1, B], [B, a]) = ([q_2, a], [*, a], R)$
2. $\delta([q_2, a], [B, b]) = ([q_2, a], [B, b], R)$
3. $\delta([q_2, a], [B, c]) = ([q_3, a], [B, c], R)$
4. $\delta([q_3, a], [*, b]) = ([q_3, a], [*, b], R)$
5. $\delta([q_3, a], [B, a]) = ([q_4, B], [*, a], L)$
6. $\delta([q_4, B], [*, a]) = ([q_4, B], [*, a], L)$
7. $\delta([q_4, B], [B, c]) = ([q_5, B], [B, c], L)$
8. $\delta([q_5, B], [B, a]) = ([q_6, B], [B, a], L)$
9. $\delta([q_6, B], [B, a]) = ([q_6, B], [B, a], L)$
10. $\delta([q_6, B], [*, a]) = ([q_1, B], [*, a], R)$
11. $\delta([q_5, B], [*, a]) = ([q_7, B], [*, a], R)$
12. $\delta([q_7, B], [B, c]) = ([q_8, B], [B, c], R)$
13. $\delta([q_8, B], [*, a]) = ([q_8, B], [*, a], R)$
14. $\delta([q_8, B], [B, B]) = ([q_9, B], [B, B], R)$

Before

State: $[q_4, B]$

*	*	*	B	*	*	*
0	1	1	c	0	1	1

.....

.....

After

State: $[q_4, B]$

*	*	*	B	*	*	*
0	1	1	c	0	1	1

1. $\delta([q_1, B], [B, a]) = ([q_2, a], [*, a], R)$
2. $\delta([q_2, a], [B, b]) = ([q_2, a], [B, b], R)$
3. $\delta([q_2, a], [B, c]) = ([q_3, a], [B, c], R)$
4. $\delta([q_3, a], [*, b]) = ([q_3, a], [*, b], R)$
5. $\delta([q_3, a], [B, a]) = ([q_4, B], [*, a], L)$
6. $\delta([q_4, B], [*, a]) = ([q_4, B], [*, a], L)$
7. $\delta([q_4, B], [B, c]) = ([q_5, B], [B, c], L)$
8. $\delta([q_5, B], [B, a]) = ([q_6, B], [B, a], L)$
9. $\delta([q_6, B], [B, a]) = ([q_6, B], [B, a], L)$
10. $\delta([q_6, B], [*, a]) = ([q_1, B], [*, a], R)$
11. $\delta([q_5, B], [*, a]) = ([q_7, B], [*, a], R)$
12. $\delta([q_7, B], [B, c]) = ([q_8, B], [B, c], R)$
13. $\delta([q_8, B], [*, a]) = ([q_8, B], [*, a], R)$
14. $\delta([q_8, B], [B, B]) = ([q_9, B], [B, B], R)$

Before

State: $[q_4, B]$

*	*	*	B	*	*	*
0	1	1	c	0	1	1

Encounters c , switches to q_5 and continues left

After

State: $[q_5, B]$

*	*	*	B	*	*	*
0	1	1	c	0	1	1

1. $\delta([q_1, B], [B, a]) = ([q_2, a], [*, a], R)$
2. $\delta([q_2, a], [B, b]) = ([q_2, a], [B, b], R)$
3. $\delta([q_2, a], [B, c]) = ([q_3, a], [B, c], R)$
4. $\delta([q_3, a], [*, b]) = ([q_3, a], [*, b], R)$
5. $\delta([q_3, a], [B, a]) = ([q_4, B], [*, a], L)$
6. $\delta([q_4, B], [*, a]) = ([q_4, B], [*, a], L)$
- 7. $\delta([q_4, B], [B, c]) = ([q_5, B], [B, c], L)$**
8. $\delta([q_5, B], [B, a]) = ([q_6, B], [B, a], L)$
9. $\delta([q_6, B], [B, a]) = ([q_6, B], [B, a], L)$
10. $\delta([q_6, B], [*, a]) = ([q_1, B], [*, a], R)$
11. $\delta([q_5, B], [*, a]) = ([q_7, B], [*, a], R)$
12. $\delta([q_7, B], [B, c]) = ([q_8, B], [B, c], R)$
13. $\delta([q_8, B], [*, a]) = ([q_8, B], [*, a], R)$
14. $\delta([q_8, B], [B, B]) = ([q_9, B], [B, B], R)$

Before

State: $[q_5, B]$

*	*	*	B	*	*	*
0	1	1	c	0	1	1

Just moved left from c and find a checked symbol. Have already considered the entire block of 0's and 1's to the left of the c . Must make sure that all the 0's and 1's to the right of c are also checked. Start moving right again, entering state q_7

After

State: $[q_7, B]$

*	*	*	B	*	*	*

1. $\delta([q_1, B], [B, a]) = ([q_2, a], [*, a], R)$
2. $\delta([q_2, a], [B, b]) = ([q_2, a], [B, b], R)$
3. $\delta([q_2, a], [B, c]) = ([q_3, a], [B, c], R)$
4. $\delta([q_3, a], [*, b]) = ([q_3, a], [*, b], R)$
5. $\delta([q_3, a], [B, a]) = ([q_4, B], [*, a], L)$
6. $\delta([q_4, B], [*, a]) = ([q_4, B], [*, a], L)$
7. $\delta([q_4, B], [B, c]) = ([q_5, B], [B, c], L)$
8. $\delta([q_5, B], [B, a]) = ([q_6, B], [B, a], L)$
9. $\delta([q_6, B], [B, a]) = ([q_6, B], [B, a], L)$
10. $\delta([q_6, B], [*, a]) = ([q_1, B], [*, a], R)$
11. $\delta([q_5, B], [*, a]) = ([q_7, B], [*, a], R)$
12. $\delta([q_7, B], [B, c]) = ([q_8, B], [B, c], R)$
13. $\delta([q_8, B], [*, a]) = ([q_8, B], [*, a], R)$
14. $\delta([q_8, B], [B, B]) = ([q_9, B], [B, B], R)$

1. $\delta([q_1, B], [B, a]) = ([q_2, a], [*, a], R)$
2. $\delta([q_2, a], [B, b]) = ([q_2, a], [B, b], R)$
3. $\delta([q_2, a], [B, c]) = ([q_3, a], [B, c], R)$
4. $\delta([q_3, a], [*, b]) = ([q_3, a], [*, b], R)$
5. $\delta([q_3, a], [B, a]) = ([q_4, B], [*, a], L)$
6. $\delta([q_4, B], [*, a]) = ([q_4, B], [*, a], L)$
7. $\delta([q_4, B], [B, c]) = ([q_5, B], [B, c], L)$
8. $\delta([q_5, B], [B, a]) = ([q_6, B], [B, a], L)$
9. $\delta([q_6, B], [B, a]) = ([q_6, B], [B, a], L)$
10. $\delta([q_6, B], [*, a]) = ([q_1, B], [*, a], R)$
11. $\delta([q_5, B], [*, a]) = ([q_7, B], [*, a], R)$
12. $\delta([q_7, B], [B, c]) = ([q_8, B], [B, c], R)$
13. $\delta([q_8, B], [*, a]) = ([q_8, B], [*, a], R)$
14. $\delta([q_8, B], [B, B]) = ([q_9, B], [B, B], R)$

Before

State: $[q_7, B]$

*	*	*	B	*	*	*
0	1	1	c	0	1	1

In state q_7 we shall surely see the c .
We enter state q_8 as we do so, and
proceed right

After

State: $[q_8, B]$

*	*	*	B	*	*	*
0	1	1	c	0	1	1

Before

State: $[q_8, B]$

*	*	*	B	*	*	*
0	1	1	c	0	1	1

M moves right in state q_8 , skipping over any checked 0's or 1's that it finds

After

State: $[q_8, B]$

*	*	*	B	*	*	*
0	1	1	c	0	1	1

1. $\delta([q_1, B], [B, a]) = ([q_2, a], [*, a], R)$
2. $\delta([q_2, a], [B, b]) = ([q_2, a], [B, b], R)$
3. $\delta([q_2, a], [B, c]) = ([q_3, a], [B, c], R)$
4. $\delta([q_3, a], [*, b]) = ([q_3, a], [*, b], R)$
5. $\delta([q_3, a], [B, a]) = ([q_4, B], [*, a], L)$
6. $\delta([q_4, B], [*, a]) = ([q_4, B], [*, a], L)$
7. $\delta([q_4, B], [B, c]) = ([q_5, B], [B, c], L)$
8. $\delta([q_5, B], [B, a]) = ([q_6, B], [B, a], L)$
9. $\delta([q_6, B], [B, a]) = ([q_6, B], [B, a], L)$
10. $\delta([q_6, B], [*, a]) = ([q_1, B], [*, a], R)$
11. $\delta([q_5, B], [*, a]) = ([q_7, B], [*, a], R)$
12. $\delta([q_7, B], [B, c]) = ([q_8, B], [B, c], R)$
13. $\delta([q_8, B], [*, a]) = ([q_8, B], [*, a], R)$
14. $\delta([q_8, B], [B, B]) = ([q_9, B], [B, B], R)$

Before

State: $[q_8, B]$

*	*	*	B	*	*	*
0	1	1	c	0	1	1

M moves right in state q_8 , skipping over any checked 0's or 1's that it finds

After

State: $[q_8, B]$

*	*	*	B	*	*	*
0	1	1	c	0	1	1

1. $\delta([q_1, B], [B, a]) = ([q_2, a], [*, a], R)$
2. $\delta([q_2, a], [B, b]) = ([q_2, a], [B, b], R)$
3. $\delta([q_2, a], [B, c]) = ([q_3, a], [B, c], R)$
4. $\delta([q_3, a], [*, b]) = ([q_3, a], [*, b], R)$
5. $\delta([q_3, a], [B, a]) = ([q_4, B], [*, a], L)$
6. $\delta([q_4, B], [*, a]) = ([q_4, B], [*, a], L)$
7. $\delta([q_4, B], [B, c]) = ([q_5, B], [B, c], L)$
8. $\delta([q_5, B], [B, a]) = ([q_6, B], [B, a], L)$
9. $\delta([q_6, B], [B, a]) = ([q_6, B], [B, a], L)$
10. $\delta([q_6, B], [*, a]) = ([q_1, B], [*, a], R)$
11. $\delta([q_5, B], [*, a]) = ([q_7, B], [*, a], R)$
12. $\delta([q_7, B], [B, c]) = ([q_8, B], [B, c], R)$
13. $\delta([q_8, B], [*, a]) = ([q_8, B], [*, a], R)$
14. $\delta([q_8, B], [B, B]) = ([q_9, B], [B, B], R)$

1. $\delta([q_1, B], [B, a]) = ([q_2, a], [*, a], R)$
2. $\delta([q_2, a], [B, b]) = ([q_2, a], [B, b], R)$
3. $\delta([q_2, a], [B, c]) = ([q_3, a], [B, c], R)$
4. $\delta([q_3, a], [*, b]) = ([q_3, a], [*, b], R)$
5. $\delta([q_3, a], [B, a]) = ([q_4, B], [*, a], L)$
6. $\delta([q_4, B], [*, a]) = ([q_4, B], [*, a], L)$
7. $\delta([q_4, B], [B, c]) = ([q_5, B], [B, c], L)$
8. $\delta([q_5, B], [B, a]) = ([q_6, B], [B, a], L)$
9. $\delta([q_6, B], [B, a]) = ([q_6, B], [B, a], L)$
10. $\delta([q_6, B], [*, a]) = ([q_1, B], [*, a], R)$
11. $\delta([q_5, B], [*, a]) = ([q_7, B], [*, a], R)$
12. $\delta([q_7, B], [B, c]) = ([q_8, B], [B, c], R)$
13. $\delta([q_8, B], [*, a]) = ([q_8, B], [*, a], R)$
14. $\delta([q_8, B], [B, B]) = ([q_9, B], [B, B], R)$

Before

State: $[q_8, B]$

*	*	*	B	*	*	*
0	1	1	c	0	1	1

M moves right in state q_8 , skipping over any checked 0's or 1's that it finds

After

State: $[q_8, B]$

*	*	*	B	*	*	*	B
0	1	1	c	0	1	1	B

M reaches a blank cell in state q_8 without encountering any unchecked 0 or 1, it accepts

1. $\delta([q_1, B], [B, a]) = ([q_2, a], [*, a], R)$
2. $\delta([q_2, a], [B, b]) = ([q_2, a], [B, b], R)$
3. $\delta([q_2, a], [B, c]) = ([q_3, a], [B, c], R)$
4. $\delta([q_3, a], [*, b]) = ([q_3, a], [*, b], R)$
5. $\delta([q_3, a], [B, a]) = ([q_4, B], [*, a], L)$
6. $\delta([q_4, B], [*, a]) = ([q_4, B], [*, a], L)$
7. $\delta([q_4, B], [B, c]) = ([q_5, B], [B, c], L)$
8. $\delta([q_5, B], [B, a]) = ([q_6, B], [B, a], L)$
9. $\delta([q_6, B], [B, a]) = ([q_6, B], [B, a], L)$
10. $\delta([q_6, B], [*, a]) = ([q_1, B], [*, a], R)$
11. $\delta([q_5, B], [*, a]) = ([q_7, B], [*, a], R)$
12. $\delta([q_7, B], [B, c]) = ([q_8, B], [B, c], R)$
13. $\delta([q_8, B], [*, a]) = ([q_8, B], [*, a], R)$
14. $\delta([q_8, B], [B, B]) = ([q_9, B], [B, B], R)$

Example

- Let's design a TM to add two binary numbers. The tape initially contains

$$w_1 c w_2 c$$

- w_1 and w_2 are two binary number (assumed to be of equal length)
- The leftmost bits are LSBs for convenience
- The c 's are separators
- We will write the sum to the right of the second c

Example-continued

1	0	1	c	0	1	1	c	B	B	B	B

Example-continued

✓											
1	0	1	c	0	1	1	c	B	B	B	B

Example-continued

✓				✓							
1	0	1	c	0	1	1	c	1	B	B	B

Example-continued

✓	✓			✓							
1	0	1	c	0	1	1	c	1	B	B	B

Example-continued

✓	✓			✓	✓						
1	0	1	c	0	1	1	c	1	1	B	B

Example-continued

✓	✓	✓		✓	✓						
1	0	1	c	0	1	1	c	1	1	B	B

Example-continued

✓	✓	✓		✓	✓	✓					
1	0	1	c	0	1	1	c	1	1	0	B

Example-continued

✓	✓	✓		✓	✓	✓					
1	0	1	c	0	1	1	c	1	1	0	1

Example-continued

- We will store two bits and the carry in the state
- States will be 4-tuples of the form $[q, a, b, c_i]$
- We start at $[q_0, B, B, 0]$
- After reading the bit from the first number, we switch to q_1 , store the bit in the state and go right
- We switch to q_2 after seeing the first c
- Once we find an unchecked bit, we switch to q_3 and store the bit in the state
- We keep going right until we see a blank. We then write the sum of the bits and the carry stored in the state

Example-continued

- We then switch to q_4 and keep going left
- We switch to q_5 when we see the first c and then to q_6 when we see the second c
- When we find the check mark we go to q_0 and move right
- Then either the cycle repeats
- Or if we see a c while at q_0 , we are done with the numbers
- In that case we switch to q_7 and keep going right until we find a blank
- We then write the carry and switch to q_8 and halt

Example-continued

 $[q_0, B, B, 0]$

1	0	1	c	0	1	1	c	B	B	B	B

Example-continued

$$[q_1, 1, B, 0]$$

✓											
1	0	1	c	0	1	1	c	B	B	B	B

Example-continued

$$[q_2, 1, B, 0]$$

✓											
1	0	1	c	0	1	1	c	B	B	B	B

Example-continued

$[q_3, 1, 0, 0]$

✓				✓							
1	0	1	c	0	1	1	c	B	B	B	B

Example-continued

 $[q_4, B, B, 0]$

✓				✓							
1	0	1	c	0	1	1	c	1	B	B	B

Example-continued

$$[q_5, B, B, 0]$$

✓				✓							
1	0	1	c	0	1	1	c	1	B	B	B

Example-continued

$$[q_6, B, B, 0]$$

✓				✓							
1	0	1	c	0	1	1	c	1	B	B	B

Example-continued

$$[q_0, B, B, 0]$$

✓				✓							
1	0	1	c	0	1	1	c	1	B	B	B

Example-continued

$$[q_0, 0, B, 0]$$

✓	✓			✓							
1	0	1	c	0	1	1	c	1	B	B	B

Example-continued

$[q_3, 0, 1, 0]$

✓	✓			✓	✓						
1	0	1	c	0	1	1	c	1	B	B	B

Example-continued

$$[q_4, B, B, 0]$$

✓	✓			✓	✓						
1	0	1	c	0	1	1	c	1	1	B	B

Example-continued

$[q_1, 1, B, 0]$

✓	✓	✓		✓	✓						
1	0	1	c	0	1	1	c	1	1	B	B

Example-continued

$[q_3, 1, 1, 0]$

✓	✓	✓		✓	✓	✓					
1	0	1	c	0	1	1	c	1	1	B	B

Example-continued

$$[q_4, B, B, 1]$$

✓	✓	✓		✓	✓	✓					
1	0	1	c	0	1	1	c	1	1	0	B

Example-continued

$$[q_0, B, B, 1]$$

✓	✓	✓		✓	✓	✓					
1	0	1	c	0	1	1	c	1	1	0	B

Example-continued

$$[q_7, B, B, 1]$$

✓	✓	✓		✓	✓	✓					
1	0	1	c	0	1	1	c	1	1	0	B

Example-continued

$$[q_8, B, B, B]$$

✓	✓	✓		✓	✓	✓					
1	0	1	c	0	1	1	c	1	1	0	1

Example-continued

- The Turing machine we shall design is:

$$M = (Q, \Sigma, \Gamma, \delta, [q_0, B, B, 0], [B, B], \{[q_7, B, B, B]\})$$

where:

Q: The set of states is

$$\{q_0, q_2, \dots, q_7\} \times \{0, 1, B\} \times \{0, 1, B\} \times \{0, 1, B\}.$$

- That is, pairs consisting of a control state q_i and three data components: 0, 1, or blank
- The data components store two bits from two numbers and a carry.
- We again use the technique of storage in the finite control, as we allow the state to remember an input symbol 0 or 1.

Example-continued

- The Turing machine we shall design is:

$$M = (Q, \Sigma, \Gamma, \delta, [q_0, B, B, 0], [B, B], \{[q_7, B, B, B]\})$$

where:

- Γ : The set of tape symbols is $\{B, *\} \times \{0, 1, c, B\}$.
- The first component, or track, can be either blank or “checked,” represented by the symbols B and $*$, respectively.
- We use the $*$ to check off symbols of the first and second groups of 0’s and 1’s.
- The second component of the tape symbol is what we think of as the tape symbol itself.
- That is, we may think of the symbol $[B, X]$ as if it were the tape symbol X , for $X = 0, 1, c, B$.

Example-continued

- The Turing machine we shall design is:

$$M = (Q, \Sigma, \Gamma, \delta, [q_0, B, B, 0], [B, B], \{[q_7, B, B, B]\})$$

where:

Σ : The input symbols are $[B, 0]$, $[B, 1]$ and $[B, c]$.

- This, as mentioned above, we identify with 0, 1 and c, respectively.

Example-continued

- The Turing machine we shall design is:

$$M = (Q, \Sigma, \Gamma, \delta, [q_0, B, B, 0], [B, B], \{[q_7, B, B, B]\})$$

where:

δ : The transition function.

Example-continued

δ is defined by the following rules, in which a , b and c_i each may stand for either 0 or 1.

1. $\delta ([q_0, B, B, c_i], [B, a]) = ([q_1, a, B, c_i], [*, a], R)$.
 - In the initial state,
 - M picks up the symbol a (which can be either 0 or 1),
 - stores it in its finite control,
 - goes to control state q_1 ,
 - “checks off” the symbol it just scanned,
 - and moves right.
 - Notice that by changing the first component of the tape symbol from B to $*$, it performs the check-off.

Example-continued

δ is defined by the following rules, in which a , b and c_i each may stand for either 0 or 1.

2. $\delta ([q_1, a, B, c_i], [B, 0/1]) = ([q_1, a, B, c_i], [B, 0/1], R)$.
- M moves right, looking for the symbol c .
 - Remember that a and b can each be either 0 or 1, independently, but can not be c .

Example-continued

δ is defined by the following rules, in which a , b and c_i each may stand for either 0 or 1.

3. $\delta ([q_1, a, B, c_i], [B, c]) = ([q_2, a, B, c_i], [B, c], R)$.
 - When M finds the c , it continues to move right, but changes to control state q_2 .

Example-continued

δ is defined by the following rules, in which a , b and c_i each may stand for either 0 or 1.

4. $\delta ([q_2, a, B, c_i], [*, 0/1]) = ([q_2, a, B, c_i], [*, 0/1], R)$.
 - In state q_2 , M continues past all checked symbols.

Example-continued

δ is defined by the following rules, in which a , b and c_i each may stand for either 0 or 1.

5. $\delta ([q_2, a, B, c_i], [B, b]) = ([q_3, a, b, c_i], [*, b], R)$.
- When it sees the first unchecked symbol, it saves the bit in the state.
 - M goes to control state q_3 and keeps moving right.

Example-continued

δ is defined by the following rules, in which a , b and c_i each may stand for either 0 or 1.

6. $\delta ([q_3, a, b, c_i], [B, 0/1/c]) = ([q_3, a, b, c_i], [B, 0/1/c], R)$.
 - It moves right over all symbols until it finds a blank

Example-continued

δ is defined by the following rules, in which a , b and c_i each may stand for either 0 or 1.

7. $\delta ([q_3, a, b, c_i], [B, B]) = ([q_4, B, B, c_o], [B, s], L)$.
- When it finds a blank, it writes the sum s to the tape and saves the carry out c_o in the state
 - s and c_o are given by

a	b	c_i	s	c_o
0	0	0	0	0
0	0	1	1	0
0	1	0	1	0
0	1	1	0	1
1	0	0	1	0
1	0	1	0	1
1	1	0	0	1
1	1	1	1	1

Example-continued

δ is defined by the following rules, in which a , b and c_i each may stand for either 0 or 1.

8. $\delta ([q_4, B, B, c_i], [B, 0/1]) = ([q_4, B, B, c_i], [B, 0/1], L)$.
- Moves over symbols until it finds c

Example-continued

δ is defined by the following rules, in which a , b and c_i each may stand for either 0 or 1.

9. $\delta ([q_4, B, B, c_i], [B, c]) = ([q_5, B, B, c_i], [B, c], L)$.
- Switches to q_5 when it sees c

Example-continued

δ is defined by the following rules, in which a , b and c_i each may stand for either 0 or 1.

10. $\delta ([q_5, B, B, c_i], [B/*, 0/1]) = ([q_5, B, B, c_i], [B/*, 0/1], L)$.
- Moves over symbols until it finds c

Example-continued

δ is defined by the following rules, in which a , b and c_i each may stand for either 0 or 1.

11. $\delta ([q_5, B, B, c_i], [B, c]) = ([q_6, B, B, c_i], [B, c], L)$.
- Switches to q_6 when it sees c

Example-continued

δ is defined by the following rules, in which a , b and c_i each may stand for either 0 or 1.

12. $\delta ([q_6, B, B, c_i], [B, 0/1]) = ([q_6, B, B, c_i], [B, 0/1], L)$.
- Moves over unchecked symbols

Example-continued

δ is defined by the following rules, in which a , b and c_i each may stand for either 0 or 1.

13. $\delta ([q_6, B, B, c_i], [*, 0/1]) = ([q_0, B, B, c_i], [*, 0/1], R)$.
- Switches to q_0 and starts moving right when it sees a checked symbol

Example-continued

δ is defined by the following rules, in which a , b and c_i each may stand for either 0 or 1.

14. $\delta ([q_0, B, B, c_i], [B, c]) = ([q_7, B, B, c_i], [B, c], R)$.
- If it sees c while at q_0 , switches to q_7

Example-continued

δ is defined by the following rules, in which a , b and c_i each may stand for either 0 or 1.

15. $\delta ([q_7, B, B, c_i], [B/*, 0/1/c]) =$
 $([q_7, B, B, c_i], [B/*, 0/1/c], R).$

- Moves over everything until it sees a blank

Example-continued

δ is defined by the following rules, in which a , b and c_i each may stand for either 0 or 1.

16. $\delta ([q_7, B, B, c_i], [B, B]) = ([q_8, B, B, B], [B, c_i], R)$.
- When it finds a blank, writes carry to tape, switches to q_8 and halts

Subroutines

- As with programs in general, it helps to think of Turing machines as built from a collection of interacting components, or “subroutines.”
- A Turing machine subroutine is a set of states that perform some useful process.
- This set of states includes a start state and another state that temporarily has no moves, and that serves as the “return” state to pass control to whatever other set of states called the subroutine.

Subroutines

- The “call” of a subroutine occurs whenever there is a transition to its initial state.
- We note that the TM has no mechanism for remembering a “return address,” that is, a state to go to after it finishes.
- We should our design of a TM call for one subroutine to be called from several states, we can make copies of the subroutine, using a new set of states for each copy.
- The “calls” are made to the start states of different copies of the subroutine, and each copy “returns” to a different state.